Self-Supervised Change Detection in Multiview Remote Sensing Images

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***Abstract*— The large amount of unlabeled remote sensing acquired from different sources and at different times (defined as multiple views in this article) presents both an opportunity and a challenge for change detection. Recently, many generative model-based methods have been proposed for remote sensing image change detection on such unlabeled data. However, the high diversities in the learned features weaken the discrimination of the relevant change indicators in unsupervised change detection tasks. Moreover, these methods lack research on massive archived images. In this work, a self-supervised change detection approach based on an unlabeled multiview setting is proposed to overcome this limitation. This is achieved by the use of a multiview contrastive loss in the feature alignment between multiview images. In this approach, a pseudo-Siamese network is trained to regress the output between its two branches pretrained in a contrastive way on a large dataset of single-sensor or cross-sensor image pairs. Finally, the feature distance between the outputs of the two branches is used to define a change measure, which can be analyzed by thresholding to get the final binary change map. Experiments are carried out on two single-sensor and three cross-sensor datasets. The proposed approach is compared with other supervised and unsupervised state-of-the-art change detection methods. Results demonstrate both improvements over state-of-the-art unsupervised methods and the proposed approach narrows the gap between unsupervised and supervised change detection.**

***Index Terms*— Change detection, multiview, remote sensing, self-supervised learning, sentinel-1/-2.**

I. Introduction

CHANGE maps are one of the most important products of remote sensing and are widely used in many applications including damage assessment and environmental monitoring. The spatial and temporal resolutions play a crucial role in obtaining accurate and timely change detection maps from multitemporal images. In this context, irrelevant changes, such as radiometric and atmospheric variations, seasonal changes of vegetation, and changes in the building shadows, which are typical of multitemporal images, limit the accuracy of change maps.

In the past decades, many researchers developed techniques that directly compare pixel values of multitemporal images to get the change maps from coarse resolution images [1]–[3], assuming that the spectral information of each pixel can completely characterize various underlying land-cover types. With the increase of spatial and spectral resolutions of remote sensing images, the use of spectral information only is often not enough to distinguish accurately land-cover changes. Many supervised and unsupervised techniques were developed to determine the land-cover changes by jointly using spatial context and spectral information. Recently, deep-learning techniques and in particular convolutional neural networks (CNNs) methods [4] have been widely used in this domain. CNNs allow one to model high-level features from images in terms of spatial and spectral information, achieving state-of-the-art results in a supervised way [5].

Most of the past works are limited to the use of single modality images. Cross-domain change detection has not received sufficient attention yet. Current Earth observation satellites provide a vast amount of multiview observations from different sensors and different times. On the one hand, images taken by different types of sensors can improve the time resolution, thus satisfying the requirement of specific applications with tight constraints. A possible example of this is the joint use of Sentinel-2 and Landsat-8 images for a regular and timely monitoring of burned areas [6]. However, the differences in acquisition modes and sensor parameters present a big challenge for traditional methods. On the other hand, multimodal data are complementary to the use of single modality images and their use becomes crucial, especially when only images from different sensors are available in some specific scenarios. This could be the case of emergency management when, for example, optical and SAR images could be jointly exploited for flood change detection tasks [7]. In this scenario, methods capable of computing change maps from images of different sensors in the minimum possible time can be very useful. This has led to the development of multisource change detection methods, which can process either multisensor or multimodal images.

Recent success of deep-learning techniques in change detection is mainly focused on supervised methods [8]–[10], which are often limited from the availability of annotated datasets. Especially, in multitemporal problems, it is expensive and often not possible to obtain a large amount of annotated samples for modeling change classes. Thus, unsupervised methods are preferred to supervised ones in many operational applications. The limited access to labeled data has driven the development of unsupervised methods, such as generative adversarial network (GAN) [11] and convolutional

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AutoEncoder (CAE) [12], which are currently among the most used deep-learning methods in unsupervised change detection tasks. Nevertheless, some studies have shown that such generative models overly focus on pixels rather than on abstract feature representations [13]. Recent researches in contrastive learning [14], [15] encourage the network to learn more interpretable and meaningful feature representations in CV tasks, where they outperformed the generative counterparts. To overcome the drawbacks of generative models, in this article, we exploit contrastive learning in multiview remote sensing image change detection.

In this work, we present a novel general approach to perform unsupervised change detection in both single-sensor and cross-sensor scenarios that is based on a multiview contrastive learning method [14]. Rather than training generative models on a predefined task, the proposed approach is trained end-to-end on large unlabeled images by minimizing the distance between features directly from bitemporal images. To this purpose, a pseudo-Siamese network (which exploits ResNet-34 as the backbone) is trained to regress the output between two branches that were pretrained by a contrastive way on a large archived multiview images. In addition, we introduce a change score that can accurately model the feature distance between bitemporal images. Changes are identified when there is a significant disagreement between the feature vectors of the two branches.

The rest of this article is organized as follows:Section II presents the related works of unsupervised change detection in multiview images including single-sensor and cross-sensor images. Section III introduces the proposed approach by describing the architecture of the pseudo-Siamese network, the considered contrastive learning approach and the change-detection method. The experimental results obtained on five different datasets and the related comparisons with supervised and unsupervised state-of-the-art methods are illustrated in Section IV. Finally, Section V draws the conclusions of the article.

II. Related Works

In the literature, unsupervised change detection techniques in multiview remote sensing images can be subdivided into two categories: single-sensor and cross-sensor. Single-sensor image change detection methods are proposed to process multitemporal images acquired by the same sensor with the same characteristics. Cross-sensor image change detection methods focus on processing heterogeneous images, which are captured by different types of sensors.

CVA [2] and its object-based variants are one of the most popular unsupervised single-sensor change detection methods. They calculate the change intensity maps and the change direction for change detection and related classification. Another popular method is the combination of PCA and K-means (PCA-KM) [16], which transforms and compares the bitemporal images in the feature space, and then determines the binary change map using k-means. In [17], Nielsen *et al.* treated the bitemporal images as multiview data and proposed the multivariate alteration detection (MAD) based on canonical correlations analysis (CCA), which maximizes the correlation between the transformed features of bitemporal images for change detection. Wu *et al.* [18] proposed a novel change detection method to project the bitemporal images into a common feature space and detected the changed pixels by extracting the invariant components based on the theory of slow feature analysis (SFA).

However, the traditional methods are easily affected by the irrelevant changes because of their weak feature representation ability in presence of high-resolution remote sensing images [19]. To get a robust feature representation, Bovolo [20] proposed a contextual parcel-based multiscale approach, which points out the usefulness of contextual information in unsupervised change detection. Object-based methods [21], [22] were also introduced to exploit spectral and spatial information effectively based on ground objects. Another popular approach to consider contextual information is based on probabilistic graphical models [23], [24], which are also used to model the complex spatial correlation between changed pixels. The development of deep-learning techniques provides new insights in jointly using spatial-contextual information. In [25], Du *et al.* developed the SFA into deep-learning methods to calculate the change intensity maps and highlight the changed components in the transformed feature space. Instead of pixel-based analysis, Saha *et al.* [4] used pretrained CNNs to extract deep spatial-spectral features from multitemporal images and analyzed the features using traditional CVA. An unsupervised learning approach, self-supervised methods (i.e., generative models and discriminative models) also are used in unsupervised change detection. Generative models [26]–[28] were adopted to generate features of bitemporal image pairs and detect the changes by an explicit comparison of the generated features. In [29], Leenstra *et al.* pretrained a discriminative model to extract features from bitemporal images on a pretext task for change detection.

Unlike single-sensor change detection, the greatest challenge in cross-sensor change detection is to align the inconsistent feature representation of different modality images. This requires transforming heterogeneous representation into a common feature space where performing change detection. There are a few traditional methods that focus on this transformation of different modalities. Gong *et al.* [30] proposed an iterative coupled dictionary-learning method that learns two coupled dictionaries for encoding bitemporal images. Luppino *et al.* [31] proposed to perform image regression by transforming images to the domain of each other and to measure the affinity matrix distance, which indicates the change possibility of each pixel. Sun *et al.* [32] developed a nonlocal patch similarity-based method by constructing a graph for each patch and establishing a connection between heterogeneous images.

Because of the ability of CNNs in feature learning, more and more techniques based on deep learning were also proposed in this area. Zhao *et al.* [33] proposed a symmetrical convolutional coupling network (SCCN) to map the discriminative features of heterogeneous images into a common feature space and generated the final change map by setting a threshold. Similarly, the conditional generative adversarial

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network (cGAN) was also used to translate two heterogeneous images into a single domain [34]. Luppino *et al.* used the change probability from [31] as the change before to guide the training of two new networks, the X-Net with two fully convolutional networks and the adversarial cyclic encoders network (ACE-Net), with two autoencoders whose code spaces are aligned by adversarial training [35]. In [36], they further jointly used domain-specific affinity matrices and autoencoders to align the related pixels from input images and reduce the impact of changed pixels.

III. Methodology

In this section, we present the proposed approach to multitemporal and multisensor remote sensing image change detection based on contrastive learning.

A. Problem Statement

Change detection is the operation of distinguishing changed and unchanged pixels of multitemporal images acquired by different sensors at different dates. Let us consider two images I_1 and I_2 acquired at two different dates t_1 and t_2, respectively. The aim of change detection is to create a change intensity map that contains permanent changes, from multiview images I_1 and I_2. As described in related works, the crucial point in this task is to align the features of unchanged pixels or patches from the different view data $T_1(\theta) = f_\theta(p_1)$ and $T_2(\phi) = g_\phi(p_2)$. Here, p_1 and p_2 are unchanged patches or pixels in images I_1 and I_2, respectively. The f and g functions are used to extract the features from multitemporal images, where θ and ϕ denote the corresponding parameters. The objective function of our task can be defined as

$$\theta, \phi = \arg\min_{\theta,\phi}\{d[f_\theta(p_1), g_\phi(p_2)]\} \tag{1}$$

where d is a measure of feature distance between T_1 and T_2.

Many change detection techniques follow this formulation including CCA, canonical information analysis (CIA), and postclassification comparison (PCC). CCA and CIA are used to calculate a linear/nonlinear relationship between features from multitemporal images. In classification-based approaches, f and g represent two classifiers trained independently or jointly [37]. Although these change detection algorithms have made some contributions to the various application scenarios, they suffer some serious drawbacks, such as the lack of robustness to the variation in data acquisition parameters. Thus, we still need the development of robust models, especially when the relevant changes are very hard to differentiate from the images. With the development of deep learning, the multiview contrastive loss was introduced in the multiview setting to get robust features. These methods are considered in this work as they can extract multiview features by maximizing the mutual information of unchanged pixels or patches between views. In Sections III-B–III-D, we will describe the proposed approach by introducing the pseudo-Siamese network, the multiview contrastive loss as well as the change detection strategy for obtaining change maps.

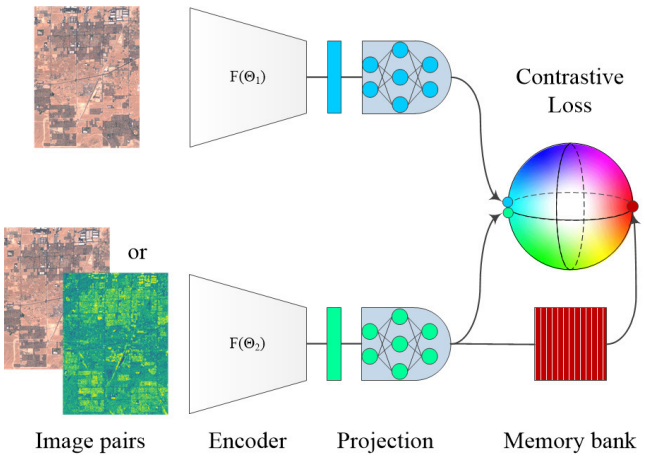


Fig. 1. Pretraining part of the proposed approach to change detection for bitemporal remote sensing image pairs. In the cross-sensor setting, the image pair consists of two images acquired by different types of sensors and the architecture of the network is symmetric with each side consisting of an encoder and a projection layer. In the single-sensor setting, the image pair consists of bitemporal images acquired by the same sensor, and two symmetric subnetworks that share almost identical architectures.

B. Pseudo-Siamese Network

Siamese networks [38] are the most used model in entities comparison. However, the comparison of cross-sensor image pairs cannot be performed by Siamese networks directly for their different imaging mechanism. Siamese networks share identical weights in two branches, while cross-sensor image pairs have dissimilar low-level features. Hence, the pseudo-Siamese network is used as the model architecture for cross-sensor image change detection. It has two branches that share the same architecture except for the input channel, but with different weights.

For single-sensor images, we propose to use the mean teacher network [39] as the architecture of our model. Mean teacher is a common pseudo-Siamese network used in self-supervised learning, which uses an exponential moving average (EMA) weight to produce a more accurate model than using the same weights directly in the single-sensor images setting. In this way, the model has a better intermediate feature representation by aggregating the information of each step.

Fig. 1 shows the architecture used in this work, where two branches are designed to extract the features of bitemporal image pairs. In this work, the ResNet-34 [40] is adopted as the backbone of the two branches and the input channels are changed for adapting to the image pairs, i.e., the polarization of SAR image patches and the spectral bands of optical images patches. In particular, we change the parameters of the strider from 2 to 1 in the third and fourth layers of the backbone for adapting the network to the relatively small input size. In greater detail, the bitemporal image pairs are passed through the unshared branches and are then modeled in output from the related feature vectors. The output feature vectors of two branches are normalized and then used to compute the similarity with each other and negative samples of the batch. Finally, the model parameters are updated by minimizing a loss function.

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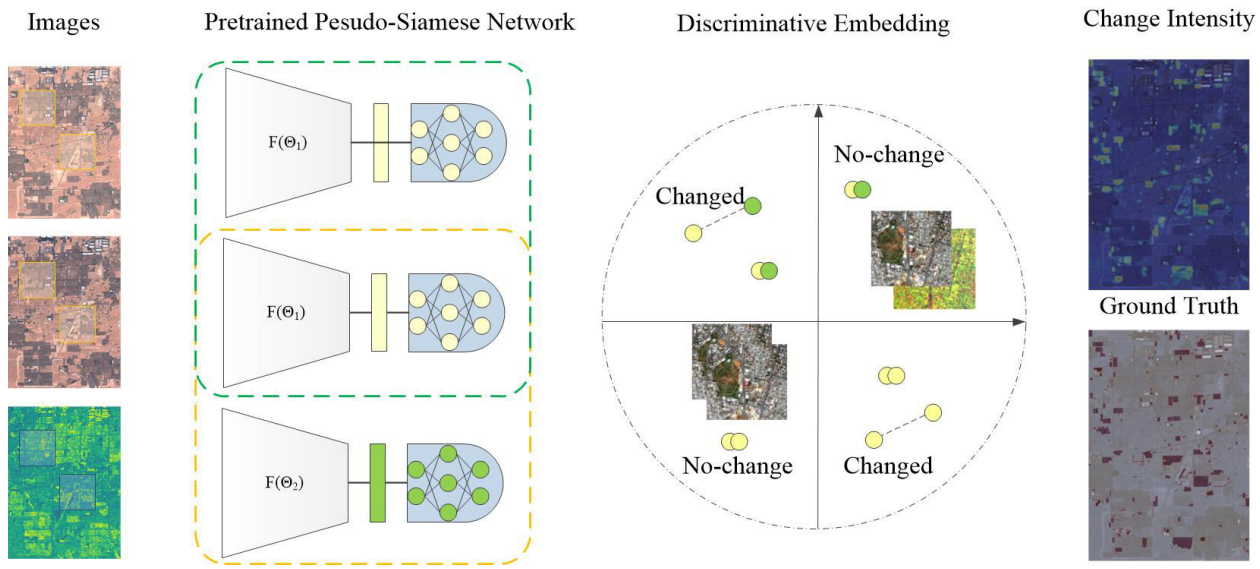


Fig. 2. Schematic overview of the proposed change detection approach (SSL). Input images are fed through the pretrained pseudo-Siamese network that extracts feature vectors from single-sensor or cross-sensor bitemporal image patches. Then, change intensity maps are generated by estimating regression errors for each pixel and the final binary change map is obtained by setting a threshold.

C. Self-Supervised Learning Approach

Contrastive learning is a popular methodology for unsupervised feature representation in the machine-learning community [14], [41]. The main idea behind the contrastive loss is to find a feature representation that attributes the feature distance between different samples. For change detection, let us consider each bitemporal image pairs $\{I_1^i, I_2^i\}_{i=1,2,\ldots,N}$ on a given scene i, which is considered as a positive pair sampled from the joint distribution $p(I_1^i, I_2^i)$. Another image pair $\{I_1^i, I_2^j\}$ taken from a different scene is considered as a negative pair sampled from the product of marginals $p(I_1^i)p(I_2^j)$. The method introduces a similarity function, $h_\theta(\cdot)$, which is used to model the feature distance between positive and negative pairs. The pseudo-Siamese network is trained to minimize the $\mathcal{L}^S_{\text{contrast}}$ defined as

$$\mathcal{L}^S_{\text{contrast}} = -\mathop{\mathbb{E}}_{S}\left[\log \frac{h_\theta(I_1^1, I_2^1)}{\sum_{j=1}^{N} h_\theta(I_1^1, I_2^j)}\right] \tag{2}$$

where (I_1^1, I_2^1) is a positive pair sample, $(I_1^1, I_2^j | j \geq 1)$ are negative pair samples, and $S = \{I_1^1, I_2^1, I_2^2, \ldots, I_2^{N-1}\}$ is a set that contains $N-1$ negative samples and one positive sample.

During the training, positive image pairs are assigned to a higher value, whereas negative pairs are assigned to a lower value. Hence, the network represents positive pairs at a close distance, whereas negative pairs at a high distance. The self-supervised method takes different augmentations of the same image as positive pairs and negative pairs sampled uniformly from different scenes. However, such a sampling strategy for negative pairs is no longer suitable in such a case. Robinson *et al.* [42] proposed an effective hard negative sampling strategy to avoid the "sampling bias" because of false-negative samples with same context information as the anchor, where anchor is the selected baseline (input) used to compare with the positive and negative inputs. With this strategy, we address the difficulty of negatives sampling in this context.

For change detection, we can construct bitemporal image sets S_1 and S_2 by fixing one set and enumerating positives and negatives from the other set. This allows us to define a symmetric loss as

$$\mathcal{L}(S_1, S_2) = \mathcal{L}^{S_1}_{\text{contrast}} + \mathcal{L}^{S_2}_{\text{contrast}}. \tag{3}$$

In practice, the noise-contrastive estimation [43] method is used to make a tractable computation of (3) when N is extremely large. This multiview contrastive learning approach makes the unsupervised change detection possible.

D. Change Detection

The change detection strategy described in this section is based on the feature learned by the contrastive learning method. Let $S = \{I_1, I_2, I_3, \ldots, I_n\}$ be a dataset of either single-sensor or cross-sensor multitemporal remote sensing images. Our goal is to detect changes between satellite images from different dates. As mentioned before, most changes of interest are those relevant to human activities, while the results are easily affected by irrelevant changes, such as seasonal changes. Other relevant changes are usually rare, whereas irrelevant changes are common during a long period. This means that, under this assumption, the features of relevant changes can be derived from the unchanged features. To this purpose, the models are trained to regress the features of images acquired at different dates. As shown in Fig. 2, here we use the considered contrastive learning algorithm to get features of either single-sensor or cross-sensor multitemporal images. After training, a change intensity map can be derived by assigning a score to each pixel indicating the probability of change.

During the network training, images acquired by the different sensors or at different dates are treated as two views in our approach. Image patches centered at each pixel are fed in

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input to the network, and the output is a single feature vector for each patch-sized input. In detail, given an input image $\mathbf{I} \in \mathbb{R}^{w \times h}$ of width w, height h, we can get a feature vector $T(r, c)$ of a square local image region with a side length p for each image pixel at row r and column c. During the inference, the model provides as output a feature map that is generated by a given size of input image patches. Let $T_1(r, c)$ and $T_2(r, c)$ denote the feature vectors at the row r and column c for the considered bitemporal images. The change intensity map is defined as the pair-wise regression error $e(r, c)$ between the feature vectors of bitemporal images

$$e(r, c) = \|T_1(r, c) - T_2(r, c)\|_2^2. \tag{4}$$

It is worth noting that the proposed model allows the use of different input sizes. To prevent the possible degeneration of the detection accuracy at a given input size, we recommend using a larger input size when the trial on a smaller input size fails.

One can see from Fig. 2 that pixels can be classified as changed and unchanged by thresholding the feature distance in the change intensity map. In this case, two strategies are considered. The simplest strategy is to choose the opposite minimum value of standardized intensity maps as the threshold value. An alternative strategy is the Robin thresholding method [44], which is robust and suitable for long-tailed distribution curves. In this method, the threshold value is the "corner" on the distribution curve of the intensity map and the maximum deviation from the straight line drawn between the endpoints of the curve. In our technique, the threshold value is determined by the first strategy if the absolute difference of these two threshold values is smaller than half of their average value. Otherwise, the threshold value is determined by the Robin thresholding method.

IV. Experimental Results

In this section, we first present the considered datasets, then the state-of-the-art change detection methods used in the comparison, and finally conduct a thorough analysis of the performance of different approaches and the analysis of robustness and efficiency.

A. Description of Datasets

We developed our experiments on five different datasets including two single-sensor datasets and three cross-sensor datasets. All remote sensing images in this work are raw images from the Google Earth Engine (GEE) and without any specific preprocessing.

1) OSCD_S2S2/_S1S1/_S1S2/_L8S2: The Onera satellite change detection (OSCD) dataset [45] was created for bitemporal change detection using Sentinel-2 images acquired between 2015 and 2018. These images have a total of 13 bands with a relatively high resolution (10 m) for visible (VIS) and near-infrared (NIR) band images and 60-m resolution for other spectral channels. The images of this dataset include urban areas and present the change type of urban growth and changes. The dataset consists of 24 pairs of multispectral images and the corresponding pixel-wise ground truth acquired in different cities and including different landscapes. The pixel-wise ground truth labels, which were manually annotated, focus on urban growth and built-up changes and contain some errors on the identification of bare lands. At the original supervised setting, 14 pairs were selected for the training set and the rest 10 pairs were used to evaluate the performance of methods.

To use this dataset in self-supervised training, we downloaded additional Sentinel-2 images in the same location as the original bitemporal images between 2016 and 2020. We considered images from each month to augment existing image pairs. Similarly, Landsat-8 multitemporal images and Sentinel-1 ground range-detected (GRD) image products are also provided in this dataset corresponding to the given Sentinel-2 scenes. The Landsat-8 images have nine channels covering the spectrum from deep blue to shortwave infrared and two long-wave infrared channels and their resolution range from 15 m to 100 m. The Sentinel-1 GRD products have been terrain corrected, multilooked, and transformed to the ground range and geographical coordinates. They consist of two channels including vertical–horizontal (VH) and vertical–vertical (VV) polarization as well as of additional information on the incidence angle.

To use this dataset for multiview change detection, we separate it into four subdatasets: OSCD_S2S2 (225 pairs), OSCD_S1S1 (577 pairs), OSCD_S1S2 (334 pairs), and OSCD_L8S2 (156 pairs). These datasets are composed of single-sensor images (OSCD_S2S2, OSCD_S1S1) and cross-sensor images (OSCD_L8S2, OSCD_S1S2). To keep consistency with previous research, ten image pairs of these four datasets corresponding to the OSCD test image pairs are treated as the test set to evaluate the performance of different methods, and image pairs acquired on other scenes and on each month of four years are used for the self-supervised pretraining. In practice, it is impossible to acquire the test image pairs of OSCD_S1S1, OSCD_L8S2, and OSCD_S1S2 at the same time as the OSCD_S2S2. Hence, we only obtained these image pairs at the closest time to OSCD_S2S2 test image pairs.

2) Flood in California: The California dataset is also a cross-sensor dataset that includes a Landsat-8 (multispectral) and a Sentinel-1 GRD (SAR) image. The multispectral and SAR images are acquired on January 5, 2017, and February 18, 2017, respectively. The dataset represents a flood occurred in Sacramento County, Yuba County, and Sutter County, California. The ground truth was extracted from a Sentinel-1 SAR image pair where the preevent image is acquired approximately at the same time as the Landsat-8 image. However, we realized that the ground truth in [31] contains many mistakes. Hence, we updated the reference data with the PCC method according to bitemporal Sentinel-1 images. The other three image pairs of Sentinel-1 and Landsat-8 images of the same scene acquired in 2017 and 2018, respectively, were used for the self-supervised pretraining of the proposed approach.

B. Experimental Settings

1) Literature Methods for Comparison: We considered different state-of-the-art methods for comparisons with the

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proposed approach on the five datasets mentioned above. On the first two optical datasets (OSCD_S2S2 and OSCD_L8S2), the proposed SSL approach was compared with two unsupervised deep-learning approaches (DSFA [25] and CAA [36]) and two deep-supervised methods (FC-EF [8] and FC-EF-Res [46]).

Code-aligned autoencoders (CAA) is a deep unsupervised methodology to align the code spaces of two autoencoders based on affinity information extracted from the multimodal input data. It allows achieving a latent space entanglement even when the input images contain changes by decreasing the interference of changed pixels. However, it degrades its performance when only one input channel is considered. It is also well suited for single-sensor change detection, as it does not depend on any prior knowledge of the data. Fully convolutional early fusion (FC-EF) is considered for the supervised change detection method on the OSCD dataset. In this method, the bitemporal image pair are stacked together as the input. The architecture of FC-EF is based on U-Net [47], where the skip connections between encoder and decoder help to localize the spatial information more precisely and get clear change boundaries. FC-EF-Res is an extension of FC-EF with residual blocks to improve the accuracy of change results. In addition, it is worth noting that the first dataset (OSCD_S2S2) has previously been extensively used in other works. Hence, we also compare our results with those of some conventional methods [45] (Log-ratio, GLRT, and Image difference), an unsupervised deep-learning method (ACGAN [48]), and supervised deep-learning techniques (FC-Siam-conc and FC-Siam-diff [45]) reported in previous articles.

On the Sentinel-1 SAR images dataset, only unsupervised methods (DSFA, SCCN, and CAA) are used for comparison. Note that some change information present in multispectral images is not detectable in SAR images, hence we did not use supervised methods on them. On the two cross-sensor remote sensing image datasets (OSCD_S1S2 and California), two state-of-the-art methods are used for comparisons, including the SCCN and CAA. Considering that only significant changes in the backscattering of SAR images can be detected, we only consider the LasVegas site in the OSCD_S1S2 dataset.

2) Implementation Details: During the training on a single-sensor dataset, we randomly composed all images acquired at adjacent months into pairs as the input. Although SAR/multispectral image pairs acquired in the same month have been used as input pairs for the rest of the multisensor dataset. After finishing the training process, the test image pairs are fed into the pretrained network and then the related change intensity maps are derived. For the supervised method (FC-EF and FC-EF-Res), we used the 14 bitemporal training images considered in the previous work [46]. In the self-supervised and supervised methods, we use four channels (VIS and NIR) in Landsat-8 and Sentinel-2 images, whereas we use two polarizations (VH and VV) in Sentinel-1 images. CAA and SCCN methods require cross-sensor image pairs having the same number of input channels. Accordingly, to keep consistency with the four input channels of multispectral images, we augmented Sentinel-1 images with the plus and minus operation between the two polarizations as the other two channels.

3) Evaluation Criteria: To appraise the different methods presented above, five evaluation metrics [precision (Pre), recall (Rec), overall accuracy (OA), F1 score, and Cohen's kappa score (Kap)] are used in this article. We simply classify the image pixels into two classes by setting an appropriate threshold value according to the presented strategy and analyze them with reference to the ground truth map. Then, the number of unchanged pixels incorrectly flagged as change is denoted by FP (false positive) and the number of changed pixels incorrectly flagged as unchanged is denoted by FN (false negative). In addition, the number of changed pixels correctly detected as change is denoted by TP (true positive) and the number of unchanged pixels correctly detected as unchanged is denoted by TN (true negative). From these four quantities, the five evaluation metrics can be defined as

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{Rec} = \frac{\text{TP}}{TP + \text{FN}} \tag{6}$$

$$F_1 = \frac{2\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}} \tag{7}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{8}$$

$$\text{Kap} = \frac{\text{OA} - \text{PE}}{1 - \text{PE}} \tag{9}$$

$$\text{PE} = \frac{(\text{TP} + \text{FP}) \cdot (\text{TP} + \text{FN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} + \frac{(\text{FN} + \text{TN}) \cdot (\text{FP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{10}$$

Obviously, a higher value of preresults in fewer false alarms, and a higher value of Rec, represents a smaller rate of incorrect detections. The overall accuracy OA is the ratio between correctly detected pixels and all pixels of the image. However, these three metrics will give a misleading overestimate of the result when the amount of changed pixels is a small fraction of the image. $F1$ score and Kap can overcome the problem of Pre and Rec and better reveal the overall performance. Note that large $F1$ and Kap values represent better overall performance.

C. Results on Single-Sensor Datasets

We first evaluate the change detection performance of the proposed approach and state-of-the-art methods (DSFA, CAA, and supervised methods) applied to the single-sensor change detection scenario. This includes bitemporal Sentinel-2 images (OSCD_S2S2 test dataset) and bitemporal Sentinel-1 images (OSCD_S1S1 test dataset). The performance metrics obtained on the OSCD_S2S2 test dataset are reported in Table I. As expected the FC-EF and FC-EF-Res supervised methods applied to raw images achieved a better performance than most unsupervised methods except the proposed SSL approach on most metrics. Among all unsupervised methods, the proposed SSL approach (with the input size of 8 pixels) with an OA of 93 % and a Kappa coefficient of 0.48, obtained the best performance on all five metrics and the best performance among

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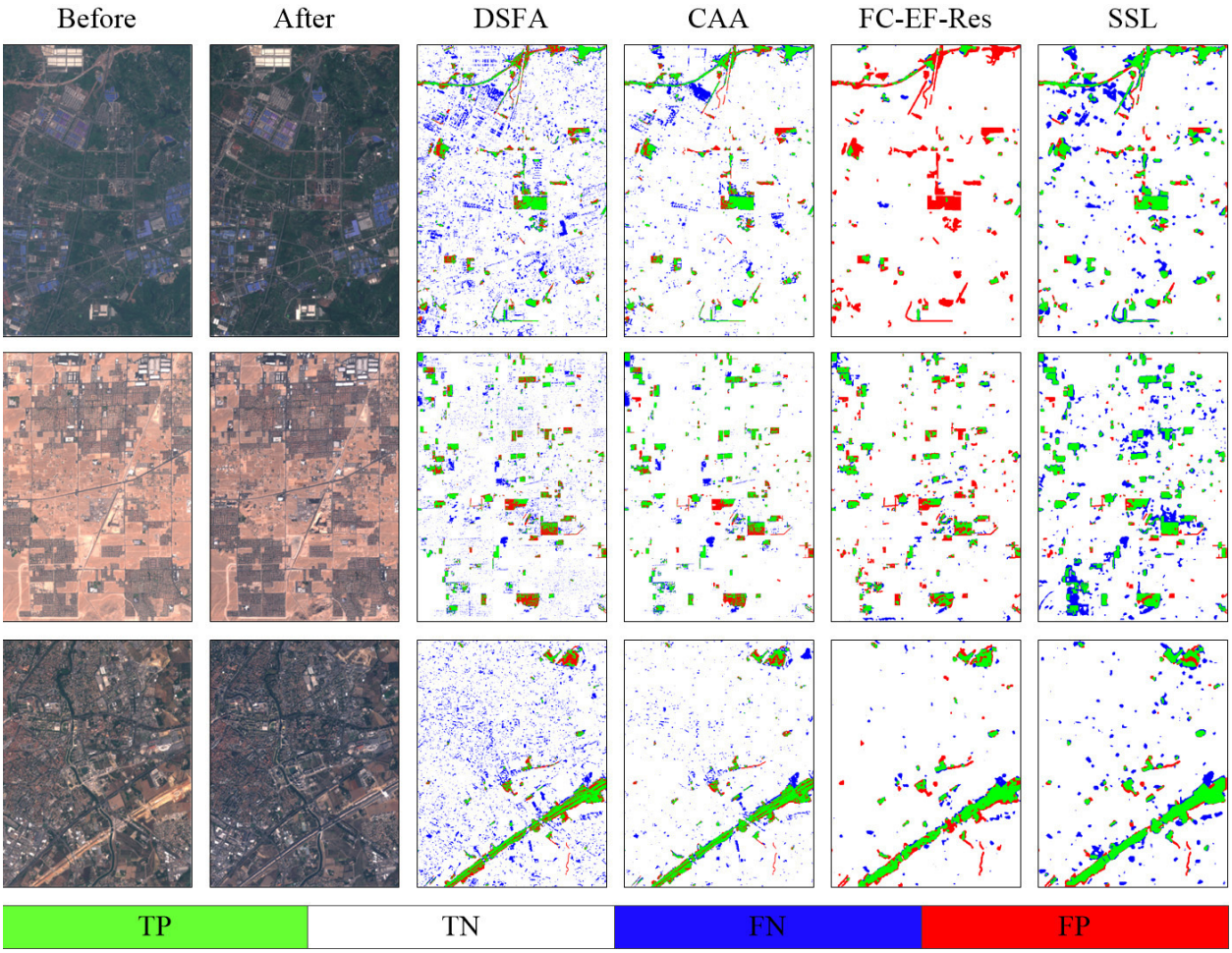


Fig. 3. Examples of change detection results on OSCD_S2S2, organized in one row for each location. Col. 1: pre-event image (Sentinel-2); Col. 2: postevent image (Sentinel-2). Change maps obtained by: DSFA (Col. 3), CAA (Col. 4), FC-EF-Res (Col. 5), and the proposed SSL (Col. 6).

all methods (included the supervised ones) implemented in this work. Although two supervised methods performed better than other methods on most metrics, they have worse performance on Recall than the proposed SSL approach. In addition, the results of other unsupervised methods (i.e., ACGAN, Image difference, GLRT, and Log-ratio) and supervised methods (i.e., Siamese and EF) on VIS and NIR channels in [45] are reported in the table. They are all worse than those of the proposed SSL approach. The results of other supervised methods (i.e., FC-EF*, FC-EF-Res*, FC-Siamese-Con*, and FC-Siamese-Diff*) applied to carefully processed RGB channel images are reported in the last rows of Table I. Their accuracies on most metrics are slightly better than those of the proposed SSL approach, but they cannot be achieved when working on raw images as a high registration precision is required. Indeed, in the related articles, multitemporal images are carefully co-registered using GEFolki toolbox to improve the accuracy of change maps [45]. On the contrary, the proposed SSL approach is based on image patches, where the registration precision of the Sentinel system is enough for obtaining a good change map.

Besides the quantitative analysis, we also provide a visual qualitative comparison in Fig. 3, where the TP, TN, FN, and FP pixels are colored in green, white, blue and red, respectively. One can see that change maps provided by DSFA and CAA are affected by a significant salt-and-pepper noise where plenty of unchanged buildings are misclassified as changed ones. This is because of the lack of use of spatial context information in these methods. This issue is well addressed by the proposed

TABLE I

QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE OSCD_S2S2 DATASET

Type	Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
Unsupervised	Prop. SSL	**40.44**	**69.10**	**93.00**	**0.51**	**0.48**
	DSFA	26.77	54.24	92.63	0.36	0.32
	CAA	23.49	52.96	91.66	0.33	0.29
	ACGAN[44]	-	64.63	77.67	-	-
	Img. Diff[41]	-	63.42	76.12	-	-
	GLRT[41]	-	60.48	76.25	-	-
	Log-ratio[41]	-	59.68	76.93	-	-
Supervised	FC-EF	**55.34**	39.48	95.13	0.46	**0.44**
	FC-EF-res	54.97	38.39	95.10	0.45	0.43
	Siamese[41]	21.57	79.40	76.76	0.34	-
	EF[41]	21.56	**82.14**	83.63	0.34	-
	FC-EF*[42]	44.72	53.92	94.23	0.49	-
	FC-EF-Res*[42]	52.27	68.24	**95.34**	**0.59**	-
	FC-Siamese-Con*[42]	42.89	47.77	94.07	0.45	-
	FC-Siamese-Diff*[42]	49.81	47.94	94.86	0.49	-

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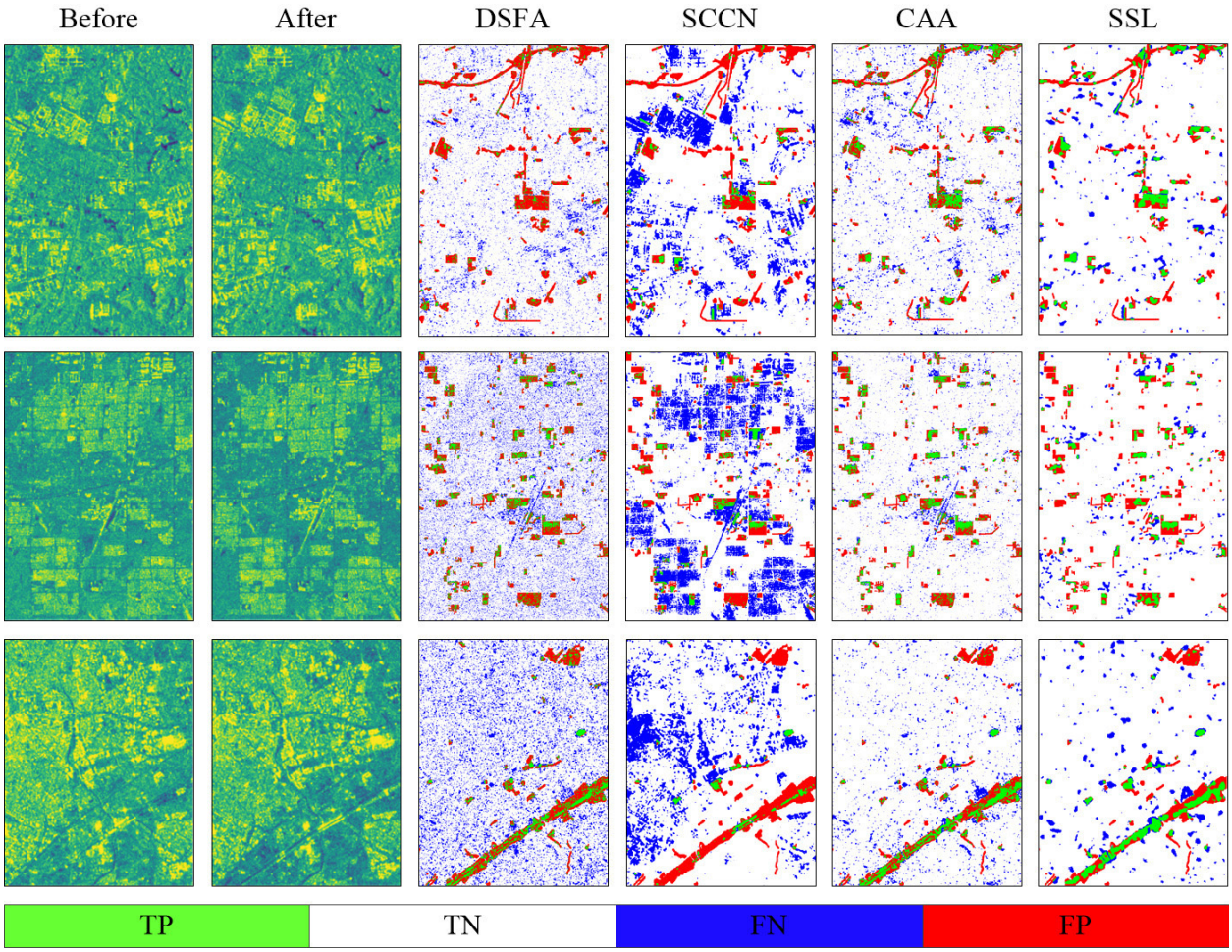


Fig. 4. Examples of change detection results on OSCD_S1S1, organized in one row for each location. Col. 1: pre-event image (Sentine-1); Col. 2: postevent image (Sentine-1). Change maps obtained by: DSFA (Col. 3), SCCN (Col. 4), CAA (Col. 5), and the proposed SSL (Col. 6).

SSL approach and the FC-EF-Res supervised method, which provides better maps. Most of the changed pixels are correctly detected in the proposed SSL approach, but with more false alarms than in the supervised FC-EF-Res method. Note that this is probably because of some small changes that are ignored in the ground truth. Nonetheless, because these results are processed in patches, some small objects are not classified correctly and false alarms on boundaries of buildings are provided by the proposed SSL approach. Instead, the change maps obtained by the FC-EF-Res method are in general more accurate and less noisy because of the use of spatial-spectral information in U-Net and the supervised learning algorithm. However, the FC-EF-Res method failed to detect most of changed pixels in the first scenario. This confirms that the change detection results of supervised methods heavily rely on the change-type distribution and the quality of training samples. This is not an issue for the proposed SSL approach.

To complete the evaluation on single-sensor datasets, the performance of all unsupervised methods is validated on the OSCD_S1S1 test dataset. The quantitative results are reported in Table II, which shows that the proposed SSL approach (with the input size of 8 pixels) produces better accuracy than other methods. The binary change maps obtained by each unsupervised method are shown in Fig. 4. One can see that all results appear much noisier because of the influence of speckle in SAR images. It is worth noting that only a new building that appeared in the postevent SAR image can be detected because minor growth of the building does not cause significant backscatter change. Apart from this, the boundaries of the detected objects are not accurate as those in the optical dataset because of the side-looking imaging mechanism. In general, the above two experiments based on single-sensor images demonstrate that the proposed SSL approach obtained the best quantitative and qualitative performance with respect to all the other considered unsupervised change detection techniques.

TABLE II
QUANTITATIVE EVALUATIONS OF DIFFERENT UNSUPERVISED APPROACHES APPLIED TO THE OSCD_S1S1 DATASETS

Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
Prop. SSL	**27.52**	27.72	92.33	**0.28**	**0.24**
SCCN	7.48	27.80	78.04	0.12	0.04
CAA	19.80	**34.81**	89.12	0.25	0.20
DSFA	10.96	22.78	**92.63**	0.15	0.08

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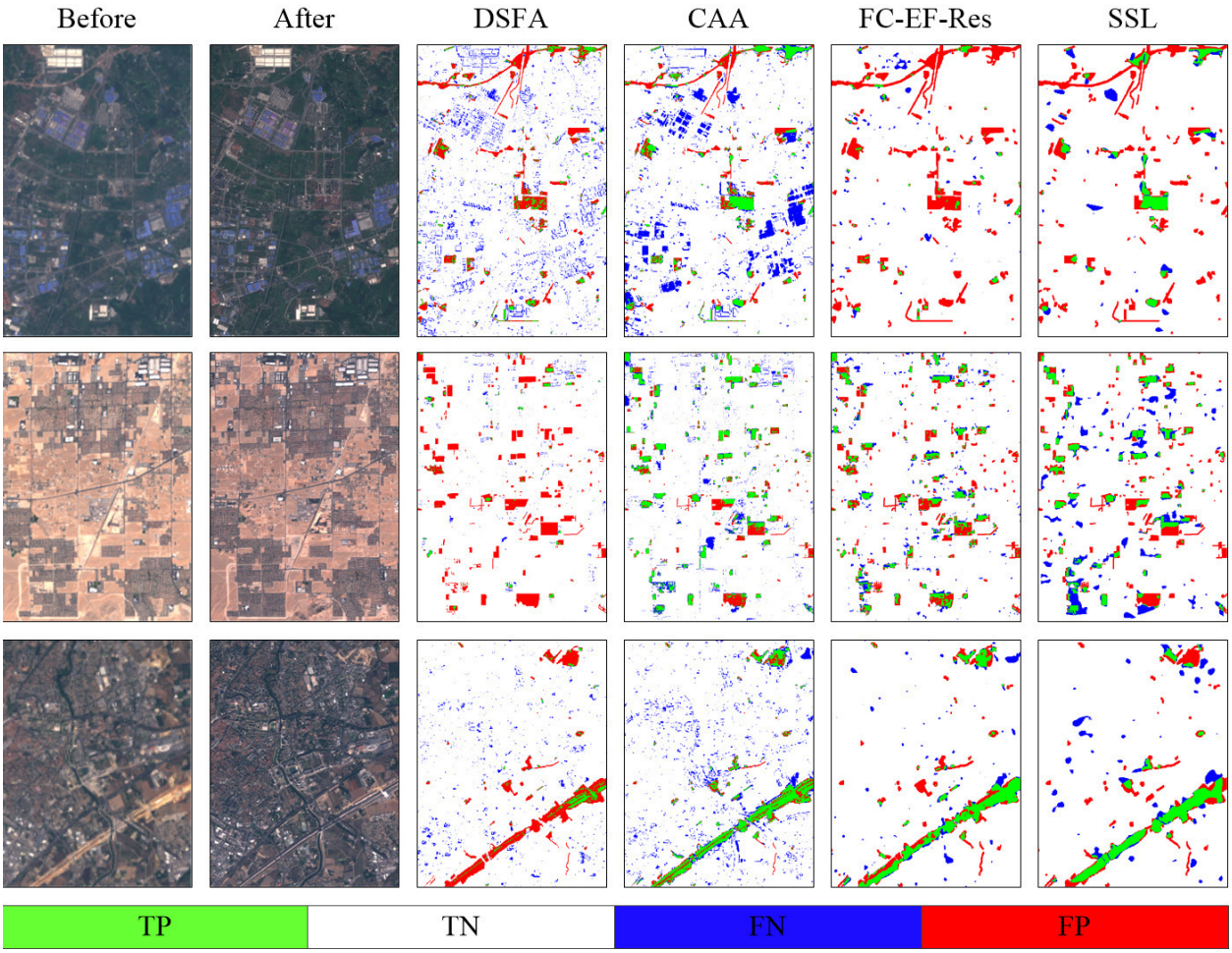


Fig. 5. Examples of change detection results on OSCD_L8S2, organized in one row for each location. Col. 1: pre-event image (Landsat-8); Col. 2: postevent image (Sentinel-2). Change maps obtained by: DSFA (Col. 3), CAA (Col. 4), FC-EF-Res (Col. 5), and the proposed SSL (Col. 6).

TABLE III
QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE OSCD_L8S2 DATASET

Type	Method	Pre(%)	Rec(%)	OA(%)	F1	Kap
Unsup.	Prop. SSL	**37.31**	32.22	**93.57**	**0.35**	**0.31**
	CAA	18.45	**45.80**	90.25	0.26	0.22
	DSFA	8.08	24.29	86.64	0.12	0.07
Sup.	FC-EF	29.75	**34.08**	92.27	**0.32**	0.28
	FC-EF-res	**39.14**	27.14	**93.93**	**0.32**	**0.29**

D. Results on Cross-Sensor Datasets

In the second change detection scenario, we consider three cross-sensor datasets, which consist of a Landsat-8/ Sentinel-2 images set (OSCD_L8S2 test dataset), a Sentinel-1/ Sentinel-2 image pair (OSCD_S1S2), and a Sentinel-1/ Landsat-8 image pair (California).

The performance of each model applied to OSCD_S2S2 is also validated on the OSCD_L8S2 test dataset, which was obtained by different optical sensors having different spatial resolutions, and the quantitative evaluation is reported in Table III. In general, the supervised methods outperform DSFA and CAA considering all five metrics. However, the performance of FC-EF-res on Recall is much worse than those of CAA and the proposed SSL approach. Meanwhile, the proposed SSL approach (with the input size of 16 pixels) with an overall accuracy of 93.57% and a Kappa coefficient of 0.31, obtained the best accuracy among methods. Fig. 5 presents the binary change maps obtained by all methods on the OSCD_L8S2. One can see that the change maps contain a larger number of false alarms for all methods compared with the maps obtained on the OSCD_S2S2. This is probably because of the relatively lower resolution of Landsat-8 VIS and NIR channel images with respect to the counterparts in Sentinel-2 images. Consistently with the results obtained on OSCD_S2S2 (see Fig. 3), the proposed SSL approach has a better segmentation result but with lower accuracy on all metrics, which indicates that the different resolution images increase the difficulty of change detection tasks.

The performance of three unsupervised methods (SCCN, CAA, and SSL) on OSCD_S1S2 is reported in Table IV. One can see that the proposed SSL approach (with the input size of 16 pixels) performs much better than the other two unsupervised methods on most metrics because of the separated training on the archived images. In contrast, SCCN and CAA are both trained on the test image only and the complicated background in the scene makes them hard to separate the unchanged pixels for the network training causing too many false alarms in change detection maps. Compared with the

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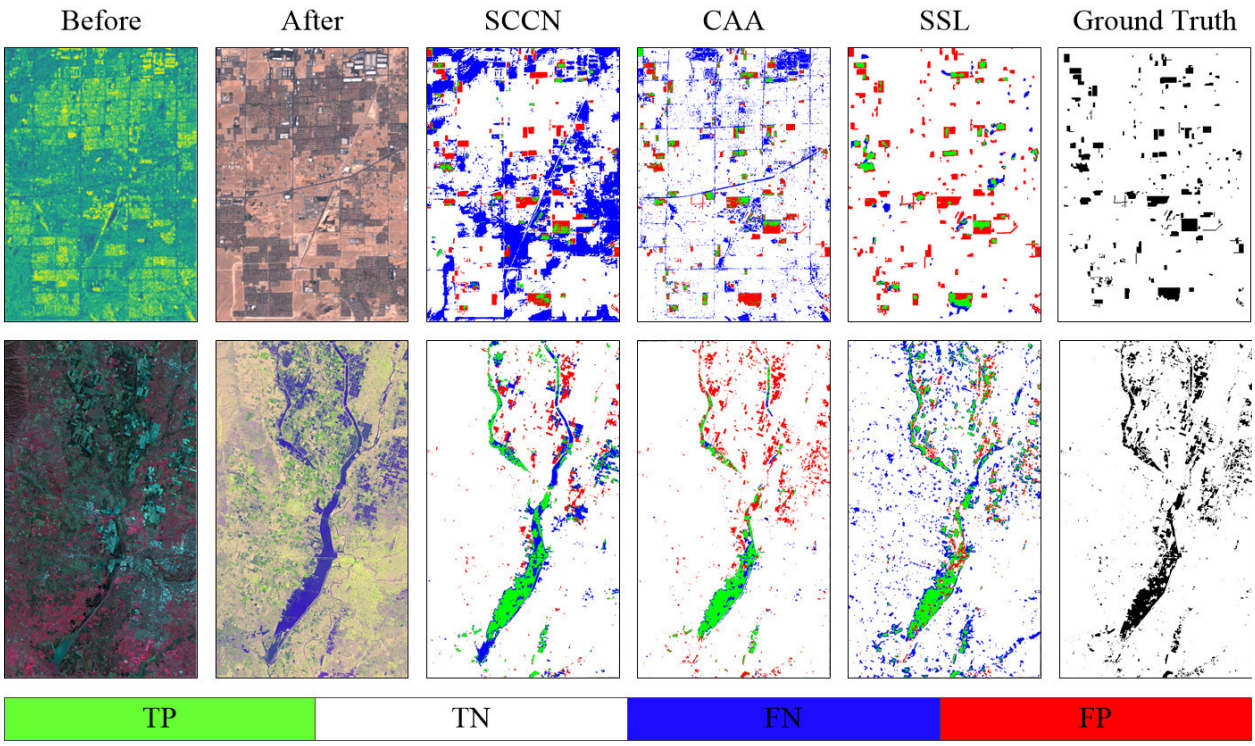


Fig. 6. Change detection results on OSCD_S1S2 and California flood, organized in one row for each location. Col. 1: pre-event image (Sentine-1 for OSCD_S1S2 and Landsat-8 for the California flood); Col. 2: post-event image (Sentine-2 for OSCD_S1S2 and Sentine-1 for the California flood). Change maps obtained by SCCN (Col. 3), CAA (Col. 4), and the proposed SSL (Col. 5). Col. 6: the ground truth.

TABLE IV
QUANTITATIVE EVALUATIONS OF DIFFERENT APPROACHES APPLIED TO THE HETEROGENEOUS IMAGES OSCD_S1S2 AND THE CALIFORNIA DATASETS

Dataset	Method	Prec(%)	Rec(%)	OA(%)	F1	Kap
S1S2	SCCN	7.38	22.45	68.54	0.11	-
	CAA	21.91	**28.71**	84.79	0.25	0.17
	Prop. SSL	**62.82**	24.19	**92.10**	**0.35**	**0.32**
California	SCCN	51.42	64.44	92.88	**0.57**	**0.53**
	CAA	**76.49**	40.38	**94.68**	0.53	0.50
	Prop. SSL	40.43	**68.14**	90.24	0.51	0.46

results obtained in the optical image experiments, the results presented here are much worse. This demonstrates the difficulty of SAR-optical change detection in complicated backgrounds, such as an urban area. Fig. 6 presents the qualitative visual results in terms of binary change maps. One can observe that the results provided by SCCN and CAA are affected by many more missed detections and false alarms than in the single-sensor case. The result of the proposed SSL approach has fewer false alarms but with more missed detections with respect to the single-sensor setting owing to the larger domain discrepancy.

Differently from the previous dataset, the California dataset is related to a simpler background and to more significant changes resulting from the flood. Table IV presents the results of all methods on this dataset. The three unsupervised methods (SCCN, CAA, and SSL) have similar performance on overall evaluation metrics (OA, F1, and Kappa). The SCCN achieves the best F1 score, Kappa, and the second-best values on Precision, Recall, and OA, whereas the CAA achieved the highest Recall. The proposed SSL approach (with the input size of 8 pixels) gets the third-best values on four of five metrics, thus it does not show obvious superiority. Fig. 6 illustrates the Landsat 8 and Sentinel-1 images and the change maps from the compared methods. Maps provided by SCCN and ACC show a clear boundary of change areas, whereas the one of the proposed SSL approach is less precise. The map of the proposed SSL approach contains more false alarms, whereas the map of the CAA has more missed detections. In general, considering the results on the two SAR-optical test datasets, the proposed SSL approach achieved the best performance in urban areas, whereas a slightly worse performance on flood detection.

E. Analysis of Robustness and Efficiency

To better analyze the robustness of the proposed SSL approach, we further evaluated the performance in terms of the five metrics considered under different input sizes. Here, we provide an example of results considering the OSCD_L8S2 dataset. We consider the effects of varying input sizes from small to large, where each input size is a multiple of 8 pixels. Table V lists that the optimal input size for the contrastive method is 16: too small (8) or too large input size (24) sharply degenerates the accuracy. The architecture of the proposed approach allows for generalization to an arbitrary input size, which can prevent failure under a given input size.

We also compared the efficiency between the proposed approach and the other selected methods in terms of inference time (Table VI). From a general perspective, it is not possible to provide a fair efficiency comparison between them.

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TABLE V
QUANTITATIVE EVALUATIONS OF CONTRASTIVE METHOD APPLIED TO OSCD_L8S2 UNDER DIFFERENT INPUT SIZES

Input Size(pixels)	Pre(%)	Rec(%)	OA(%)	F1	Kappa
8	17	10.73	92.52	0.13	0.09
16	**37.31**	**32.22**	**93.57**	**0.35**	**0.31**
24	11.8	11.19	90.9	0.11	0.07

TABLE VI
EFFICIENCY COMPARISONS BETWEEN DIFFERENT METHODS

Models	CAA	DSFA	SCCN	FC-EF	FC-EF-res	Prop. SSL
Kappa	0.29	0.32	0.06	0.44	0.43	**0.48**
Time (s)	**0.003**	0.09	0.001	0.015	0.015	7.849

This is because they are not end-to-end deep-learning models. Because the supervised method does not allow arbitrary input size and memory limitation, we used a small patch size $(1 \times 8 \times 512 \times 512)$ that fits in memory to test inference time for each model. The times reported for all models are based on the use of PyTorch on a 7.8 GB RTX 2070ti GPU.

From the analysis of Table VI, one can see that the unsupervised models (CAA, DSFA, and SCCN) need much less inference time but result in far low accuracy. The proposed approach needs more inference time. However, this is acceptable at an operational level when working with a proper GPU setting. Note that a key performance indicator to consider is that, when compared with the supervised methods, the proposed approach achieved much higher accuracy and without any label during the network training. We expect that the time cost of annotating labels in supervised methods can be higher than the inference time in the proposed approach. Moreover, the proposed approach also has a parameter redundancy and the tradeoff between accuracy and inference time depends on the particular task considered.

V. CONCLUSION

In this work, we have presented a self-supervised approach to unsupervised change detection in multiview remote sensing images, which can be used with both single-sensor and cross-sensor images. The main idea of the presented framework is to extract good feature representations from multiview images using contrastive learning. Images from satellite mission archives are used to train the pseudo-Siamese network without using any label. Under the reasonable assumption that the change event is rare in long-time archived images, the network can properly align the features learned from images obtained at different times even when they contain changes. After completing the pretraining process, the regression error of image patches captured from bitemporal images can be used as a change score to indicate the change probability. If required, a binary change map can be directly calculated from change intensity maps by using a thresholding method.

Experimental results on both single-sensor and cross-sensor remote sensing image datasets proved that the proposed SSL approach can be applicable in practice, and demonstrated its superiority over several state-of-the-art unsupervised methods. Results also show that the performance declines when the resolution of the two sensors is different in a homogeneous setting. Moreover, in the SAR-optical change detection setting, the change detection results are affected by the complexity of the background.

As a final remark, note that in this work we only considered bitemporal images to detect changes. This has negative impacts on false alarms. Our future work will be focused on the refinement of changed maps by further decreasing false alarms by combining a larger number of images from the time-series.

REFERENCES

[1] L. Bruzzone and S. B. Serpico, "Detection of changes in remotely-sensed images by the selective use of multi-spectral information," *Int. J. Remote Sens.*, vol. 18, no. 18, pp. 3883–3888, Dec. 1997.

[2] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.

[3] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[4] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[5] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[6] D. P. Roy *et al.*, "Landsat-8 and Sentinel-2 burned area mapping—A combined sensor multi-temporal change detection approach," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111254.

[7] M. Huang and S. Jin, "Rapid flood mapping and evaluation with a supervised classifier and change detection in Shouguang using Sentinel-1 SAR and Sentinel-2 optical data," *Remote Sens.*, vol. 12, no. 13, p. 2073, Jun. 2020.

[8] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[9] F. Rahman, B. Vasu, J. Van Cor, J. Kerekes, and A. Savakis, "Siamese network with multi-level features for patch-based change detection in satellite imagery," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 958–962.

[10] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[11] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[12] J. Masci, U. Meier, D. Cireşan, and J. Schmidhuber, "Stacked convolutional auto-encoders for hierarchical feature extraction," in *Proc. Int. Conf. Artif. Neural Netw.* Berlin, Germany: Springer, 2011, pp. 52–59.

[13] X. Liu *et al.*, "Self-supervised learning: Generative or contrastive," vol. 1, no. 2, 2020, *arXiv:2006.08218*. [Online]. Available: http://arxiv.org/abs/2006.08218

[14] Y. Tian, D. Krishnan, and P. Isola, "Contrastive multiview coding," in *Computer Vision—ECCV 2020*. Cham, Switzerland: Springer, 2020, pp. 776–794.

[15] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 9729–9738.

[16] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[17] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, Apr. 1998.

[18] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[19] L. Bruzzone and F. Bovolo, "A novel framework for the design of change-detection systems for very-high-resolution remote sensing images," *Proc. IEEE*, vol. 101, no. 3, pp. 609–630, Mar. 2013.

[20] F. Bovolo, "A multilevel parcel-based approach to change detection in very high resolution multitemporal images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 1, pp. 33–37, Jan. 2009.

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[21] G. Chen, G. J. Hay, L. M. T. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.
[22] L. Ma *et al.*, "Object-based change detection in urban areas: The effects of segmentation strategy, scale, and feature space on unsupervised methods," *Remote Sens.*, vol. 8, no. 9, p. 761, Sep. 2016.
[23] C. Benedek and T. Szirányi, "Change detection in optical aerial images by a multilayer conditional mixed Markov model," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3416–3430, Oct. 2009.
[24] G. Cao, L. Zhou, and Y. Li, "A new change-detection method in high-resolution remote sensing images based on a conditional random field model," *Int. J. Remote Sens.*, vol. 37, no. 5, pp. 1173–1189, Mar. 2016.
[25] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.
[26] N. Lv, C. Chen, T. Qiu, and A. K. Sangaiah, "Deep learning and superpixel feature extraction based on contractive autoencoder for change detection in SAR images," *IEEE Trans. Ind. Informat.*, vol. 14, no. 12, pp. 5530–5538, Dec. 2018.
[27] C. Ren, X. Wang, J. Gao, X. Zhou, and H. Chen, "Unsupervised change detection in satellite images with generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, early access, Dec. 31, 2021, doi: 10.1109/TGRS.2020.3043766.
[28] H. Dong, W. Ma, Y. Wu, J. Zhang, and L. Jiao, "Self-supervised representation learning for remote sensing image change detection based on temporal prediction," *Remote Sens.*, vol. 12, no. 11, p. 1868, Jun. 2020.
[29] M. Leenstra, D. Marcos, F. Bovolo, and D. Tuia, "Self-supervised pretraining enhances change detection in Sentinel-2 imagery," in *Pattern Recognition. ICPR International Workshops and Challenges*. Cham, Switzerland: Springer, 2021, pp. 578–590.
[30] M. Gong, P. Zhang, L. Su, and J. Liu, "Coupled dictionary learning for change detection from multisource data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7077–7091, Dec. 2016.
[31] L. T. Luppino, F. M. Bianchi, G. Moser, and S. N. Anfinsen, "Unsupervised image regression for heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9960–9975, Dec. 2019.
[32] Y. Sun, L. Lei, X. Li, H. Sun, and G. Kuang, "Nonlocal patch similarity based heterogeneous remote sensing change detection," *Pattern Recognit.*, vol. 109, Jan. 2021, Art. no. 107598.
[33] W. Zhao, Z. Wang, M. Gong, and J. Liu, "Discriminative feature learning for unsupervised change detection in heterogeneous images based on a coupled neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7066–7080, Dec. 2017.
[34] X. Niu, M. Gong, T. Zhan, and Y. Yang, "A conditional adversarial network for change detection in heterogeneous images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 45–49, Jan. 2019.
[35] L. T. Luppino *et al.*, "Deep image translation with an affinity-based change prior for unsupervised multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Feb. 17, 2021, doi: 10.1109/TGRS.2021.3056196.
[36] L. T. Luppino *et al.*, "Code-aligned autoencoders for unsupervised change detection in multimodal remote sensing images," 2020, *arXiv:2004.07011*. [Online]. Available: http://arxiv.org/abs/2004.07011
[37] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.
[38] J. Bromley, Y. LeCun, C. Moore, E. Säckinger, and R. Shah, "Signature verification using a 'Siamese' time delay neural network," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 7, no. 4, pp. 669–688, 1993.
[39] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1195–1204.
[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[41] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*. [Online]. Available: http://arxiv.org/abs/1807.03748
[42] J. Robinson, C.-Y. Chuang, S. Sra, and S. Jegelka, "Contrastive learning with hard negative samples," in *Advances in Neural Information Processing Systems*, vol. 33. Red Hook, NY, USA: Curran Associates, 2020, pp. 21798–21809.
[43] M. Gutmann and A. Hyvärinen, "Noise-contrastive estimation: A new estimation principle for unnormalized statistical models," in *Proc. 13th Int. Conf. Artif. Intell. Statist., JMLR Workshop Conf.*, 2010, pp. 297–304.
[44] P. L. Rosin and J. Hervás, "Remote sensing image thresholding methods for determining landslide activity," *Int. J. Remote Sens.*, vol. 26, no. 6, pp. 1075–1092, Mar. 2005.
[45] R. C. Daudt, B. L. Saux, A. Boulch, and Y. Gousseau, "Urban change detection for multispectral Earth observation using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2115–2118.
[46] R. C. Daudt, B. L. Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.
[47] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.
[48] S. Saha, Y. T. Solano-Correa, F. Bovolo, and L. Bruzzone, "Unsupervised deep learning based change detection in Sentinel-2 images," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Aug. 2019, pp. 1–4.



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